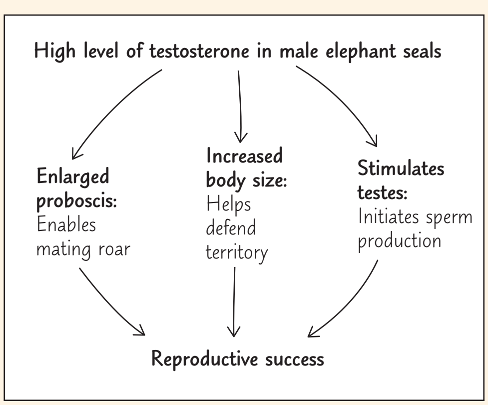
45 Hormones and the Endocrine System

Figure 45.1 Male and female elephant seals (*Mirounga angustirostris*) differ greatly in appearance and behavior. The male is much larger, and only he has the prominent proboscis for which the species is named. The male is also far more territorial, using the proboscis to emit loud roars during mating season. Underlying each of these differences is a single hormone—testosterone. Like all hormones, testosterone is an endocrine signaling molecule that circulates in the blood throughout the body.

KEY CONCEPTS

Study Tip

Make a flowchart: Many hormones, such as insulin, parathyroid hormone, and epinephrine, have multiple physiological effects in a single organism. To keep track of the action and function of each such hormone, make a flowchart like this example. Use arrows to indicate how the hormone's diverse effects contribute to an overall outcome for the organism.



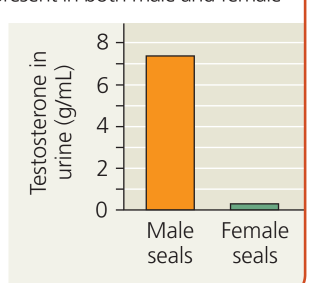
Go to Mastering Biology

For Students (in eText and Study Area)
- Get Ready for Chapter 45
- Animation: Water-Soluble Hormone Pathway
- Animation: Steroid Hormone Pathway

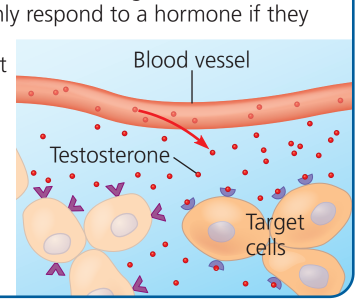
For Instructors to Assign (in Item Library)
- Scientific Skills Exercise: Designing a Controlled Experiment
- Problem-Solving Exercise: Is Thyroid Regulation Normal in this Patient?

What variables shape a hormone's effect on an animal's body and behavior?

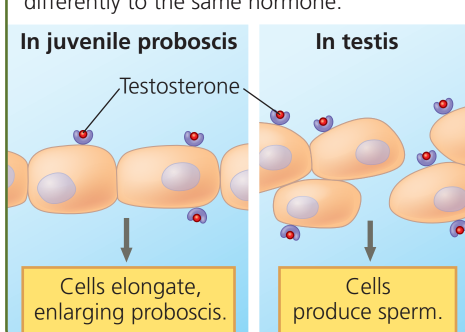
Concentration of the hormone in the body:
Testosterone is present in both male and female mammals, but typically at a much higher concentration in males.



Presence of the hormone receptor in a cell:
A hormone circulates throughout the bloodstream, but cells only respond to a hormone if they have a receptor that binds that hormone specifically.



Response of the cell when the receptor binds the hormone:
Cells in different tissues may respond differently to the same hormone.



▼ **Male elephant seals sparring**